Project: Worked examples
Title: Column base plate (EN 1993-1-8)
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## 1. Scope and method

An unstiffened rectangular base plate under an I/H, RHS or CHS column on grouted concrete: bearing resistance, the force regime and holding-down bolt tension for a major-axis moment, the tension T-stub of the plate and the compression T-stub, shear by friction and anchor bolts, and the plate thickness that satisfies them together. The bolt tension and shear per anchor go to Hilti PROFIS for the anchorage, which is done outside this sheet.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Bearing (cl. 6.2.5) | $f_{cd} = \alpha_{cc} \cdot f_{ck} / \gamma_c$; $f_{jd} = \beta_j \cdot \alpha \cdot f_{cd}$, $\beta_j = 2/3$; $c = t_p \cdot \sqrt{[f_y / (3 \cdot f_{jd} \cdot \gamma_{M0})]}$ |
| Axial (cl. 6.2.8.2) | $N_{j,Rd} = f_{jd} \cdot A_{eff}$, $A_{eff}$ the bearing strips of width c round the column footprint (full rectangle when they overlap); $u_N = N_{c,Ed} / N_{j,Rd}$ |
| Moment (cl. 6.2.8.3, Table 6.7) | $z_C$ from the section ($(h - t_f)/2$, $(h - t)/2$, half-annulus centroid) or entered, $x_b$ the bolt row from the axis, $z = z_C + x_b$; $C_{Ed} = (M_{Ed} + N_{Ed} \cdot x_b)/z$ and $T_{Ed} = C_{Ed} - N_{Ed}$ when both are ≥ 0; with $N_{Ed} > 0$ and $T_{Ed} < 0$ both sides are in compression, $C = N_{Ed}/2 \pm M_{Ed}/(2z_C)$; with $N_{Ed} < 0$ and $C_{Ed} < 0$ both rows are in tension, $T = -N_{Ed}/2 \pm M_{Ed}/(2x_b)$; $F_{t,Ed} = T_{Ed}/n_t$ at the more loaded ro… |
| Plate T-stub (cl. 6.2.6.11, 6.2.4, Table 6.2) | $M_{pl,Rd} = 0.25 \cdot l_{eff} \cdot t_p^2 \cdot f_y / \gamma_{M0}$; without prying $\min(2 \cdot M_{pl,Rd}/m, n_t \cdot F_{t,Rd})$; with prying the modes $4 \cdot M_{pl,Rd}/m$, $(2 \cdot M_{pl,Rd} + n \cdot \Sigma F_{t,Rd})/(m + n)$ with $n = \min(e_{min}, 1.25m)$, and $n_t \cdot F_{t,Rd}$; $l_{eff}$ is the total for the $n_t$ bolts |
| Compression T-stub (cl. 6.2.6.9, 6.2.6.10) | $C_{Rd} = f_{jd} \cdot A_{flange}$, checked when $M_{Ed} > 0$ and the resultant is in compression |
| Shear (cl. 6.2.2) | $F_{2,vb,Rd} = \alpha_{bc} \cdot f_{ub} \cdot A_s / \gamma_{M2}$, $\alpha_{bc} = 0.44 - 0.0003 \cdot f_{yb}$; $F_{vb,Rd} = \min(F_{1,vb,Rd}, F_{2,vb,Rd})$; $V_{Rd} = C_{f,d} \cdot N_{c,Ed} + n_b \cdot F_{vb,Rd}$; $V_{b,Ed} = \max(V_{Ed} - C_{f,d} \cdot N_{c,Ed}, 0)/n_b$ |
| Required thickness | the smallest $t_p$ in 10–100 mm with $u_N$, $u_C$ and $u_T$ all ≤ 1, $z_C$, $C_{Ed}$ and $T_{Ed}$ re-evaluated at every trial $t_p$ |

Basis: Design resistance of column bases in shear[1](cl. 6.2.2); Equivalent T-stub in tension[1](cl. 6.2.4, Table 6.2); Equivalent T-stub in compression[1](cl. 6.2.5); Concrete in compression including grout[1](cl. 6.2.6.9); Base plate in bending under compression[1](cl. 6.2.6.10); Base plate in bending under tension[1](cl. 6.2.6.11); Anchor bolts in tension[1](cl. 6.2.6.12); Column bases subject to axial forces[1](cl. 6.2.8.2); Column bases subject to axial force and bending moment[1](cl. 6.2.8.3, Table 6.7); Design strength of concrete; partial area loading[2](cl. 3.1.6, 6.7).

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| column | Column section type | I/H | | drawing: PF-GA-001 rev C | any |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| h | Column depth, along the plate length L (I/H and RHS) | 260.3 | mm | spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy | 50 … 1200 mm |
| b | Column width, along the plate width B (I/H and RHS) | 256.3 | mm | spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy | 50 … 1200 mm |
| $t_f$ | Flange thickness (I/H) | 17.3 | mm | spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy | 3 … 100 mm |
| $t_w$ | Web thickness (I/H) | 10.3 | mm | spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy | 3 … 100 mm |
| B | Plate width | 400 | mm | drawing: PF-BP-001 rev B | 100 … 1500 mm |
| L | Plate length | 450 | mm | drawing: PF-BP-001 rev B | 100 … 1500 mm |
| $t_p$ | Plate thickness | 25 | mm | drawing: PF-BP-001 rev B | 10 … 100 mm |
| $f_y$ | Plate yield strength | 265 | MPa | spec: BS EN 10025-2 S275, 16 mm < t ≤ 40 mm | 235 … 460 MPa |
| $f_{ck}$ | Concrete characteristic cylinder strength | 30 | MPa | spec: BS EN 206 C30/37 pad foundation, drawing PF-FN-001 | 20 … 50 MPa |
| $f_{ck,g}$ | Grout characteristic strength (optional: checks the $\beta_j$ = 2/3 condition) | 60 | MPa | spec: Non-shrink grout datasheet, 28-day strength (placeholder GR-DS-01) | 5 … 100 MPa |
| $t_g$ | Grout thickness (optional: checks the $\beta_j$ = 2/3 condition) | 30 | mm | drawing: PF-BP-001 rev B | 1 … 200 mm |
| $\alpha_{cc}$ | $\alpha_{cc}$, long-term effects on the concrete strength | 0.85 | | spec: UK NA to BS EN 1992-1-1, NA to cl. 3.1.6(1)P | 0.8 … 1 |
| $\gamma_c$ | $\gamma_c$, partial factor for concrete | 1.5 | | spec: UK NA to BS EN 1992-1-1, Table NA.1 | 1.2 … 1.8 |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| $\gamma_{M0}$ | $\gamma_{M0}$, partial factor for the plate | 1 | | spec: UK NA to BS EN 1993-1-1, NA.2.15 | 1 … 1.25 |
| $\gamma_{M2}$ | $\gamma_{M2}$, partial factor for the anchor bolts | 1.25 | | spec: UK NA to BS EN 1993-1-8, NA.2.3 | 1.1 … 1.5 |
| $C_{fd}$ | $C_{f,d}$, friction coefficient plate to grout | 0.2 | | spec: EN 1993-1-8 cl. 6.2.2, sand-cement grout | 0 … 0.45 |
| α | α, concentration factor $\sqrt{(A_{c1}/A_{c0})}$ of the foundation (1 when the foundation geometry is not known) | 1.5 | | assumption: Conservative: a 1.2 m square pad under the 0.45 × 0.40 m plate gives A_c1/A_c0 well above 2.25; checked on drawing PF-FN-001. | 1 … 3 |
| $\alpha_{basis}$ | Foundation geometry that α rests on (state it when α > 1) | Pad 1.2 × 1.2 m, 0.8 m deep (PF-FN-001): A_c1/A_c0 above 2.25, so α = 1.5 is taken | | drawing: PF-FN-001 rev A | any |
| $N_{Ed}$ | Axial force, COMPRESSION positive (the opposite sign to S3's $N_{Ed}$, which is tension positive; S2's $R1_y$, an upward reaction, is compression positive) | 210 | kN | reference: S5 load combinations sheet, ULS 6.10b, wind leading, column base A | -5000 … 10000 kN |
| $M_{Ed}$ | Major-axis moment (uniaxial) | 95 | kN·m | reference: S5 load combinations sheet / S2 frame sheet, column base A | 0 … 2000 kN·m |
| $V_{Ed}$ | Shear force | 60 | kN | reference: S5 load combinations sheet / S2 frame sheet, column base A | 0 … 2000 kN |
| $n_t$ | Holding-down bolts in each bolt row (one row each side) | 2 | | drawing: PF-BP-001 rev B | 1 … 20 |
| $n_b$ | Holding-down bolts in total (both sides) | 4 | | drawing: PF-BP-001 rev B | 2 … 40 |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| $x_b$ | Bolt row distance from the column axis (outside the column face, inside the plate edge) | 185 | mm | drawing: PF-BP-001 rev B | 50 … 1000 mm |
| m | Bolt to flange-weld toe distance, tension T-stub (required: it follows from $x_b$, the column and the weld) | 46.9 | mm | assumption: Bolt at 185 mm from the axis, flange face at 130.15 mm: 54.85 mm, less 0.8·√2 × 7 mm (a 10 mm fillet leg has a 7 mm throat) = 7.9 mm. | 5 … 300 mm |
| $l_{eff}$ | Effective length of the tension T-stub, total for the $n_t$ bolts (required: from the cl. 6.2.6.5 patterns for the plate) | 350 | mm | assumption: Total for the two bolts of the row, about 175 mm each from the cl. 6.2.6.5 patterns (engineer's estimate). | 10 … 2000 mm |
| prying | Prying forces develop (short bolts, $L_b \le L_b^*$); off is conservative for the plate | no | | assumption: Cast-in holding-down bolts: prying not assumed (min(2·M_pl/m, ΣF_t) is the conservative case). | any |
| $F_{t,Rd}$ | Tension resistance per bolt (from S4) | 203 | kN | reference: S4 bolt sheet, M24 8.8 (203.04 kN), rounded down | 1 … 2000 kN |
| $f_{ub}$ | Anchor bolt ultimate strength | 800 | MPa | spec: EN ISO 898-1 class 8.8 | 400 … 1000 MPa |
| $f_{yb}$ | Anchor bolt yield strength ($\alpha_{bc}$ is valid for 235–640 MPa) | 640 | MPa | spec: EN ISO 898-1 class 8.8 | 235 … 640 MPa |
| $A_s$ | Anchor bolt tensile stress area (from S4) | 352.5 | mm² | reference: S4 bolt sheet, M24 stress area (352.50 mm²) | 50 … 1500 mm² |
| $F_{1,vb,Rd}$ | Bolt bearing resistance on the plate, $F_{1,vb,Rd}$ (from S4 $F_{b,Rd}$) | 364.4 | kN | reference: S4 bolt sheet, M24 bearing on 25 mm S275 plate, $e_1$ = 60 mm, $d_o$ = 27 mm (364.44 kN) | 1 … 2000 kN |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| $profis_{ref}$ | Hilti PROFIS report reference (the report is filed as Evidence) | PROFIS-BP1 (to be filed) | | assumption: The PROFIS report is not attached in this example, so the sheet warns that the anchorage is not checked. | any |

## 3. Assumptions

A1. $N_{Ed}$ is positive in compression here, the opposite of S3 (tension positive): a link from S2 must take the reaction $R1_y$ (up positive, compression into the base), never the member force $N_{e1}$. (Sign convention.)
A2. Static loading; a major-axis moment only (uniaxial); an unstiffened rectangular plate on grout of adequate strength and thickness for $\beta_j$ = 2/3 (checked when the grout strength and thickness are entered). (Scope of S6.)
A3. The simplified T-stub model: one compression resultant at the centre of the compression T-stub (flange centre, or the half-annulus centroid of a CHS) and one tension resultant per bolt row; the force regime follows EN 1993-1-8 Table 6.7. The web is not given credit in the moment check. (EN 1993-1-8 cl. 6.2.8.3.)
A4. The effective length $l_{eff}$ is the total for the $n_t$ bolts of a row, read from the cl. 6.2.6.5 patterns; it is used for modes 1 and 2 alike ($l_{eff,1} \le l_{eff,2}$, so this is conservative for mode 2). (One input.)
A5. Concrete, anchor tension, pull-out, breakout, edge and the tension–shear interaction of the anchors are designed in Hilti PROFIS from $F_{t,Ed}$ and $V_{b,Ed}$ per anchor; the report is filed as Evidence and referenced. With prying switched on the prying force Q is not in $F_{t,Ed}$ and PROFIS must add it. (External anchorage design.)
A6. The friction resistance counts only the compression, $C_{f,d}\cdot\max(N_{Ed}, 0)$; a column in net tension has none. Anchor bolts share shear equally, with their holes assumed no larger than the standard clearance (oversized or slotted holes are outside this sheet). (EN 1993-1-8 cl. 6.2.2.)
A7. The bearing outstand c is cut back where it would pass the plate edge; the web and the column interior keep the full c. The CHS compression side is half of the annular bearing area, with its centroid as lever arm (an approximation). (Conservative against an uncut c.)
A8. The compression T-stub is checked against the flange area alone as soon as $M_{Ed} > 0$, while at $M_{Ed} = 0$ the whole footprint (web included) resists, so the compression check jumps at $M_{Ed} = 0^+$ (conservative). (Limitation.)
A9. Partial factors and the friction coefficient come from the records in use ($\gamma_c$, $\alpha_{cc}$: EN 1992-1-1 NA; $\gamma_{M2}$: EN 1993-1-8 NA; $\gamma_{M0}$: EN 1993-1-1 NA; $C_{f,d}$: the grout used); the values shown default to the EN recommended and UK values. (A small set of factors only; no table from a standard is shipped.)
A10. α (α, concentration factor $\sqrt{(A_{c1}/A_{c0})}$ of the foundation (1 when the foundation geometry is not known)) is taken as 1.5. (Conservative: a 1.2 m square pad under the 0.45 × 0.40 m plate gives $A_{c1}/A_{c0}$ well above 2.25; checked on drawing PF-FN-001.)
A11. m (Bolt to flange-weld toe distance, tension T-stub (required: it follows from $x_b$, the column and the weld)) is taken as 46.9 mm. (Bolt at 185 mm from the axis, flange face at 130.15 mm: 54.85 mm, less 0.8·√2 × 7 mm (a 10 mm fillet leg has a 7 mm throat) = 7.9 mm.)
A12. $l_{eff}$ (Effective length of the tension T-stub, total for the $n_t$ bolts (required: from the cl. 6.2.6.5 patterns for the plate)) is taken as 350 mm. (Total for the two bolts of the row, about 175 mm each from the cl. 6.2.6.5 patterns (engineer's estimate).)
A13. prying (Prying forces develop (short bolts, $L_b \le L_b^*$); off is conservative for the plate) is taken as no. (Cast-in holding-down bolts: prying not assumed ($\min(2\cdot M_{pl}/m, \Sigma F_t)$ is the conservative case).)
A14. $profis_{ref}$ (Hilti PROFIS report reference (the report is filed as Evidence)) is taken as PROFIS-BP1 (to be filed). (The PROFIS report is not attached in this example, so the sheet warns that the anchorage is not checked.)

## 4. Working

[31] $f_{cd} = \alpha_{cc} * f_{ck} / \gamma_c$ [2](cl. 3.1.6)
$= 0.85 \times 30\ \text{MPa} / 1.5$
$= 17\ \text{MPa}$
Concrete design compressive strength

[32] $\beta_j = 2/3$, fixed [1](cl. 6.2.5)
= grout conditions of cl. 6.2.5 assumed met
$= 0.66667$
Joint coefficient $\beta_j$

[33] $f_{jd} = \beta_j * \alpha * f_{cd}$ [1](cl. 6.2.5)
$= 0.66667 \times 1.5 \times 17\ \text{MPa}$
$= 17\ \text{MPa}$
Design bearing strength of the joint

[34] $c = t_p * \text{sqrt}(f_y / (3 * f_{jd} * \gamma_{M0}))$ [1](cl. 6.2.5)
$= 25\ \text{mm} \times \text{sqrt}(265\ \text{MPa} / (3 \times 17\ \text{MPa} \times 1))$
$= 56.987\ \text{mm}$
Additional bearing width round the column footprint

[35] $A_{eff} = 2(b + 2c)(t_f + 2c) + (h - 2t_f - 2c)(t_w + 2c)$ [1](cl. 6.2.5)
$= 2 \times 370.27 \times 131.27 + 111.73 \times 124.27\ \text{mm}^2$
$= 111100\ \text{mm}^2$
Effective bearing area

[38] $N_{j,Rd} = f_{jd} * A_{eff}$ [1](cl. 6.2.8.2, 6.2.5)
$= 17\ \text{MPa} \times 111100\ \text{mm}^2$
$= 1888.7\ \text{kN}$
Axial bearing resistance of the joint

[39] $N_{c,Ed} = \max(N_{Ed}, 0)$
$= N_{Ed} = 210\ \text{kN}$
$= 210\ \text{kN}$
Design compression in the column

[40] $u_N = N_{c,Ed} / N_{j,Rd}$
$= 210\ \text{kN} / 1888.7\ \text{kN}$
$= 0.11119$
Axial bearing utilisation

[41] $t_{req,N}$ = smallest $t_p$ in 10–100 mm with $f_{jd} \cdot A_{eff}(c(t_p)) \geq N_{c,Ed}$ (bisection) [1](cl. 6.2.5)
$= N_{c,Ed} = 210\ \text{kN}$; carried at the 10 mm minimum
$= 10\ \text{mm}$
Plate thickness required for axial bearing alone

[42] $z_C = (h - t_f)/2$ [1](cl. 6.2.8.3)
$= (260.3 - 17.3)/2\ \text{mm}$
$= 121.5\ \text{mm}$
Compression resultant distance from the column axis

[43] $z = z_C + x_b$ [1](cl. 6.2.8.3)
$= 121.5 + 185$
$= 306.5\ \text{mm}$
Lever arm between the compression resultant and the bolt row

[44] $C_{Ed} = (M_{Ed} + N_{Ed} \cdot x_b)/z$ [1](cl. 6.2.8.3, Table 6.7)
$= (95\ \text{kN·m} + 210\ \text{kN} \times 185\ \text{mm})/306.5\ \text{mm}$
$= 436.7\ \text{kN}$
Compression resultant under the compression flange

[45] $T_{Ed} = C_{Ed} - N_{Ed}$ [1](cl. 6.2.8.3, Table 6.7)
$= 436.7 - 210$
$= 226.7\ \text{kN}$
Tension resultant at the bolt row

[46] $C_{Ed,2}$ = none
= other side
= 0 kN
Compression resultant under the other flange
[47] $T_{Ed,2}$ = none
= other row
= 0 kN
Tension resultant at the other bolt row
[48] $F_{t,Ed} = T_{Ed} / n_t$ [1](cl. 6.2.6.12)
= 226.7 kN / 2
= 113.35 kN
Holding-down bolt tension per bolt at the more loaded row (→ PROFIS)
[49] $A_{flange}$ = (b + 2c)($t_f$ + 2c), at most $A_{eff}/2$ [1](cl. 6.2.6.9)
= min(370.27 × 131.27, $A_{eff}/2$) mm²
= 48608 mm²
Bearing area of the compression flange
[50] $C_{Rd} = f_{jd} * A_{flange}$ [1](cl. 6.2.6.9, 6.2.6.10)
= 17 MPa × 48608 mm²
= 826.33 kN
Bearing resistance of the compression flange
[51] $u_C = C_{Ed} / C_{Rd}$
= 436.7 kN / 826.33 kN
= 0.52849
Compression flange utilisation
[52] $M_{pl,Rd} = 0.25 * l_{eff} * t_p$^2 * $f_y / \gamma_{M0}$ [1](Table 6.2)
= 0.25 × 350 mm × (25 mm)^2 × 265 MPa / 1
= 14.492 kN·m
Plastic moment resistance of the plate over $l_{eff}$
[53] $T_{Rd,1} = 2 * M_{pl,Rd} / m$ [1](Table 6.2)
= 2 × 14.492 kN·m / 46.9 mm
= 618 kN
Mode 1-2 without prying forces
[54] $T_{Rd,3} = n_t * F_{t,Rd}$ [1](Table 6.2)
= 2 × 203 kN
= 406 kN
Mode 3: bolt failure
[55] $T_{Rd} = \min(T_{Rd,1}, T_{Rd,3})$ [1](cl. 6.2.4)
= min(618 kN, 406 kN)
= 406 kN
Tension T-stub resistance (no prying)
[56] $u_T = T_{Ed} / T_{Rd}$
= 226.7 kN / 406 kN
= 0.55839
Plate bending and bolts in tension, T-stub utilisation
[57] $u_{bt} = F_{t,Ed} / F_{t,Rd}$
= 113.35 kN / 203 kN
= 0.55839
Holding-down bolt tension utilisation
[58] $t_{req}$ = smallest $t_p$ in 10–100 mm with $u_N$, $u_C$, $u_T \le 1$, with $z_C$, $C_{Ed}$ and $T_{Ed}$ recomputed at every trial $t_p$ (scan and bisection) [1](cl. 6.2.5, 6.2.6.9, 6.2.6.11)
= $N_{c,Ed}$ = 210 kN, $C_{Ed}$ = 436.7 kN, $T_{Ed}$ = 226.7 kN
= 15.142 mm
Plate thickness required

[59] $\alpha_{bc} = 0.44 - 0.0003 \cdot f_{yb}$ [1](cl. 6.2.2)
$= 0.44 - 0.0003 \times 640$
$= 0.248$
Anchor bolt shear factor $\alpha_{bc}$
[60] $F_{2,vb,Rd} = \alpha_{bc} * f_{ub} * A_s / \gamma_{M2}$ [1](cl. 6.2.2)
$= 0.248 \times 800\ MPa \times 352.5\ mm^2 / 1.25$
$= 55.949\ kN$
Anchor bolt shear resistance $F_{2,vb,Rd}$
[61] $F_{vb,Rd} = \min(F_{1,vb,Rd}, F_{2,vb,Rd})$ [1](cl. 6.2.2)
$= \min(364.4\ kN, 55.949\ kN)$
$= 55.949\ kN$
Anchor bolt shear resistance per bolt (bearing or shear)
[62] $F_{f,Rd} = C_{fd} * N_{c,Ed}$ [1](cl. 6.2.2)
$= 0.2 \times 210\ kN$
$= 42\ kN$
Friction resistance, compression only
[63] $V_{Rd} = F_{f,Rd} + n_b * F_{vb,Rd}$ [1](cl. 6.2.2)
$= 42\ kN + 4 \times 55.949\ kN$
$= 265.8\ kN$
Shear resistance of the base
[64] $u_V = V_{Ed} / V_{Rd}$
$= 60\ kN / 265.8\ kN$
$= 0.22574$
Shear utilisation
[65] $V_{b,Ed} = \max(V_{Ed} - F_{f,Rd}, 0)/n_b$ [1](cl. 6.2.2)
$= \max(60 - 42, 0)/4$
$= 4.5\ kN$
Shear per anchor bolt after friction (→ PROFIS)
[66] $u_{max} = \max(u_N, u_C, u_T, u_{bt}, u_V)$
$= u_N = 0.11119,\ u_C = 0.52849,\ u_T = 0.55839,\ u_{bt} = 0.55839,\ u_V = 0.22574$
$= 0.55839$
Governing utilisation

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $F_{t,Ed}$ | 113.35 kN | Bolt tension per bolt (to PROFIS) |
| $V_{b,Ed}$ | 4.500 kN | Shear per anchor bolt (to PROFIS) |
| $N_{j,Rd}$ | 1888.7 kN | Axial bearing resistance |
| $u_{max}$ | 0.56 | Governing utilisation |
| $f_{cd}$ | 17 MPa | Concrete design strength |
| $\beta_j$ | 0.66667 | Joint coefficient $\beta_j$ |
| $f_{jd}$ | 17 MPa | Bearing strength $f_{jd}$ |
| c | 56.99 mm | Bearing outstand c |
| $A_{eff}$ | 111100 mm² | Effective bearing area |
| $t_{req,N}$ | 10 mm | Plate thickness required for axial bearing alone |
| $t_{req}$ | 15.14 mm | Plate thickness required (bearing, compression and tension T-stubs) |
| $z_C$ | 121.5 mm | Compression lever arm $z_C$ |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| z | 306.5 mm | Lever arm z |
| $C_{Ed}$ | 436.70 kN | Compression resultant |
| $C_{Ed,2}$ | 0 kN | Compression resultant, other side |
| $T_{Ed}$ | 226.70 kN | Tension resultant, more loaded row |
| $T_{Ed,2}$ | 0 kN | Tension resultant, other row |
| $A_{flange}$ | 48608 mm² | Compression-side bearing area |
| $C_{Rd}$ | 826.33 kN | Compression-side resistance |
| $M_{pl,Rd}$ | 14.49 kN·m | Plate plastic moment |
| $T_{Rd,1}$ | 618.00 kN | T-stub mode 1-2, no prying |
| $T_{Rd,3}$ | 406 kN | T-stub mode 3 |
| $T_{Rd}$ | 406 kN | Tension T-stub resistance |
| $\alpha_{bc}$ | 0.248 | Anchor shear factor $\alpha_{bc}$ |
| $F_{2,vb,Rd}$ | 55.95 kN | Anchor shear resistance $F_{2,vb,Rd}$ |
| $F_{vb,Rd}$ | 55.95 kN | Anchor shear resistance per bolt |
| $F_{f,Rd}$ | 42 kN | Friction resistance |
| $V_{Rd}$ | 265.80 kN | Shear resistance |
| regime | "tension at one bolt row, compression under the other flange (Table 6.7, z = $z_c$ + $x_b$)" | Force regime (Table 6.7) |
| $T_{mode}$ | "mode 3 (bolt failure)" | Governing T-stub mode |
| $u_N$ | 0.11 | Axial bearing utilisation |
| $u_C$ | 0.53 | Compression-side utilisation |
| $u_T$ | 0.56 | Tension T-stub utilisation (governing) |
| $u_{bt}$ | 0.56 | Bolt tension utilisation |
| $u_V$ | 0.23 | Shear utilisation |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Base plate tension T-stub (bending at the bolts) | $u_T$ = 0.56 | $u_T \le 1$ | 0.56 | [1](cl. 6.2.6.11) |
| PASS | Holding-down bolt tension | $u_{bt}$ = 0.56 | $u_{bt} \le 1$ | 0.56 | [1](cl. 6.2.6.12) |
| PASS | Compression flange bearing | $u_C$ = 0.53 | $u_C \le 1$ | 0.53 | [1](cl. 6.2.6.9, 6.2.6.10) |
| PASS | Shear: friction plus anchor bolts | $u_V$ = 0.23 | $u_V \le 1$ | 0.23 | [1](cl. 6.2.2) |

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Concrete bearing under the axial force | $u_N = 0.11$ | $u_N \leq 1$ | 0.11 | [1](cl. 6.2.8.2, 6.2.5) |

## 7. Applicability

WARNING records: tabulated values not taken from a reference record are $\alpha_{cc}$, $\gamma_c$ (EN 1992-1-1 National Annex factors); $\gamma_{M2}$ (EN 1993-1-8 National Annex factor); $\gamma_{M0}$ (EN 1993-1-1 National Annex factor); $C_{fd}$ (Friction coefficient plate to grout); $f_y$ (Plate strength). Enter each as one so its source, edition and licence travel with it. [record]
WARNING $f_y$: one yield strength (265 MPa) is used at every plate thickness, but EN 10025-2 lowers the yield strength of a grade as the thickness passes 16 mm and again above 40 mm. $t_p$ = 25 mm and $t_{req}$ = 15.142 mm pass the first step: enter the yield strength for the plate thickness in use (steel.strengths record), so $M_{pl,Rd}$ and the T-stub are not overstated. [$f_{y,thickness}$]
WARNING $u_{profis}$: the anchorage is not assessed on this sheet. The concrete failure modes of the anchors (cone, pull-out, edge, shear, interaction), which usually govern, are designed in PROFIS; $u_{max}$ covers the steel parts only, whether or not a report is attached. Enter the report's governing utilisation to include it. [anchorage-result]
WARNING $profis_{ref}$: 'PROFIS-BP1 (to be filed)' is typed text, not an attached report; attach the Hilti PROFIS report as Evidence so the reference points at the filed document. The anchorage design is not checked on this sheet. [anchorage]

## 8. Verification

Template struct.base-plate@1.4.0 Content hash cf631b3ae895bf39 Engine calc-engine/1.0.0
Known-answer cases passed 9/9 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

## 9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-8: Design of Joints. BS EN 1993-1-8:2005. BSI; 2005.
2. EN 1992-1-1:2004.